Probability and Statistics B - Practical Sheet 5

Getting started:

1. Open **RStudio** (do not use the base R application).
2. From the menu, select **File** → **New File** → **R Script**. This will create a blank script where you can type and run your R commands.
3. Immediately save your script with a meaningful name, e.g., `Practical1.R`, using **File** → **Save As**. Choose a location you can easily find later.
4. Type your commands in the script and run them using:
   - **Ctrl + Enter** (Windows) or
   - **Cmd + Enter** (Mac)
5. When you have your answers, copy the relevant output (not the entire script) into a Word document.

Important: Always keep your R script saved. This is good practice and ensures you have a record of your work.

1. Consider the results below that give data on highway traffic flow. It shows the relationship between traffic flow, x (in thousands of vehicles per day) and lead content, y (micrograms per gram) of the bark of nearby trees.

   ```
   x:  8.3 8.3 12.1 12.1 17.0 17.0 17.0 24.3 24.3 24.3 33.6
   y:  227 312  362  521  640  539  728  945  738  759 1263
   ```

   (a) Enter these data into two columns in R.

   (b) Plot the data using `plot(`$y \sim x$`)`, labelling the diagram appropriately.

   (c) You now wish to find the regression line of y on x. This has the equation $y = a + bx$. The values a and b can be obtained in R using the `lm` ("linear model") command `lm(y~x)`. This gives the equation for y in terms of x (or at least the gradient b and point that it cuts the y axis, a). Assign this to a vector called "regress".

   ```
   regress=lm(y ~ x)
   ```

   Write down the equation of the best straight line and copy and paste the output from

   ```
   summary(regress)
   ```

   From the summary, decide whether the values of a and b are significantly different from 0. Note: The P-values are given in the output.

   (d) The calculated straight-line equation can be drawn on the graph using

   ```
   plot(y~x)
   abline(lm(y~x))
   ```

   Copy and paste this graph into your Word document.

   (e) Find the residuals using the command

   ```
   residuals(regress)
   ```

   (Recall that the residuals are the differences between the observed value of the dependent variable y and the predicted value $\hat{y}$).

   It is often useful to look at a plot of the residuals against x. Do this using

```
plot(residuals(regress)~x)
```

Do they follow any pattern?

(f) Obtain the Product-Moment Correlation Coefficient, r, and comment on what its value means.

```
cor(x,y)
```

(g) Predict the 95% confidence interval for the expected mean value of lead content when the traffic flow is 26.0 thousands of vehicles per day

```
predict(regress,data.frame(x=26.0),interval="confidence",level=0.95)
```

(h) Predict the 95% confidence interval for an individual value of lead content when the traffic flow is 26.0 thousands of vehicles per day

```
predict(regress,data.frame(x=26.0),interval="prediction",level=0.95)
```

2. At the end of the skiing season the tourist board in a mountain region examines the records of ten ski resorts. For each one it obtains the total number (y, thousands) of visitor-days during the season as a measure of the resort's popularity, and the ski-lift capacity (x, thousands), being the maximum number of skiers that can be transported per hour. The resulting data are given in Table 1.

Resort	A	B	C	D	E	F	G	H	I	J
Lift capacity x:	1.9	3.3	1.2	4.2	1.5	2.2	1.0	5.6	1.9	3.8
Visitor-days y:	15.1	22.6	9.2	37.5	8.9	21.1	5.8	41.0	9.2	32.4

Table 1: Records of ski resorts

(a) Produce a scatterplot of the data and comment on it by making reference to the correlation coefficient, r.

(b) Choosing "Visitor-days" as the subject of the equation and assuming the usual model for linear regression, $Y_i = \alpha + \beta x_i + e_i$, use R to obtain the parameter estimates $\hat{\beta}$ and $\hat{\alpha}$. Produce another scatterplot, but this time also showing the least-squares regression line of best fit on the graph.

(c) Obtain a scatter plot of residuals against fitted values and suggest how this can be used to assess how good the regression line/equation is at modelling the data.

```
residuals(lm(y~x))
fitted.values(lm(y~x))
```

(d) Suppose that a resort can increase its ski-lift capacity by 500 skiers per hour. Estimate the increase in the number of visitor-days it can expect in a season